Nachiket Karve

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Research Interests – Nonlinear Dynamics, Chaos Theory, Statistical Mechanics

Education

Boston University
PhD, Physics *2021 – Present*
Advisor: Dr David Campbell

Indian Institute of Technology, Kanpur
BS-MS Dual Degree, Physics *2016 – 2021*
GPA: 9.6/10.0
Advisors: Dr Loganayagam R, Dr Nilay Kundu
Thesis: Non-Polynomial Divergences in Wightman Correlation Functions

Papers and Publications

[1] N. Karve, N. Rose, H. Kim, D. K. Campbell, and A. Polkovnikov. *Understanding Chaos through Physical Observables.* Manuscript in preparation.

[2] N. Rose, N. Karve, and D. K. Campbell. *Gradient of the Adiabatic Gauge Potential in Classical Systems.* 2025. arXiv: 2508.03804 [nlin.CD]. URL: https://arxiv.org/abs/2508.03804.

[3] N. Karve, N. Rose, and D. Campbell. *Diffusion as a Signature of Chaos.* 2025. arXiv: 2507.18617 [nlin.CD]. URL: https://arxiv.org/abs/2507.18617.

[4] N. Karve, N. Rose, and D. Campbell. "Adiabatic gauge potential as a tool for detecting chaos in classical systems". In: *Phys. Rev. Res.* 7 (3 July 2025), p. 033006. DOI: 10.1103/1z5x-j644. URL: https://link.aps.org/doi/10.1103/1z5x-j644.

[5] N. Karve, N. Rose, and D. Campbell. "Periodic orbits in Fermi–Pasta–Ulam–Tsingou systems". In: *Chaos: An Interdisciplinary Journal of Nonlinear Science* 34.9 (Sept. 2024), p. 093117. ISSN: 1054-1500. DOI: 10.1063/5.0223767. eprint: https://pubs.aip.org/aip/cha/article-pdf/doi/10.1063/5.0223767/20161085/093117_1_5.0223767.pdf. URL: https://doi.org/10.1063/5.0223767.

[6] N. Karve and R. Loganayagam. *Heisenberg Picture for Open Quantum Systems.* 2020. arXiv: 2011.15118 [quant-ph]. URL: https://arxiv.org/abs/2011.15118.

Talks

Chaos in Classical Systems, *Boston University, May 2025* [slides]

Chaos in FPUT-like Systems, *March Meeting 2025, Anaheim* [slides]

The Metastable State of the FPUT Problem, *March Meeting 2024, Minneapolis* [slides]

The Metastable State of the FPUT Problem, *March Meeting 2023, Las Vegas* [slides]

Awards and Honors

KVPY (Kishore Vaigyanik Protsahan Yojana) Fellowship *2016 – 2021*

Academic Excellence Award, IIT Kanpur *2016, 2017, 2018, 2019*

Silver Medal, University Physics Competition *2017*

Teaching Experience

Boston University *2021 – Present*

Quantum Computing (PY 536), Teaching Fellow and Guest Lecturer

General Physics I (PY 211), Grader

General Physics II (PY 212), Teaching Fellow and Grader

Physics I (PY 105), Teaching Fellow

Physics II (PY 106), Teaching Fellow

Mentorship

Undergraduate Students

Emir Akdag

Jonah Gluck

Kristen Bestavros

Academic Services

Physical Review Referee *2025 – Present*

Head and Volunteer at Prayas, IIT Kanpur's Community Welfare School *2018 – 2021*

Academic Mentor, IIT Kanpur *2017 – 2018*